

CHATBOT_With_LCEL_with_MessageHistory_Trim_Message

```
!pip install langgraph
!pip install langchain-google-genai
!pip install langchain-community
!pip install langchain-chroma
!pip install langchain-huggingface
```

Show hidden output

```
import warnings
warnings.filterwarnings('ignore')

from langchain_community.chat_message_histories import ChatMessageHistory
from langchain_core.chat_history import BaseChatMessageHistory
from langchain_core.runnables.history import RunnableWithMessageHistory
```

```
from google.colab import userdata
from langchain_google_genai import ChatGoogleGenerativeAI
model = ChatGoogleGenerativeAI(model="gemini-3.6-flash", temperature=0, api_key=userdata.get("GEMINI"))
```

```
store={}

def get_session_history(session_id:str)->BaseChatMessageHistory:
    if session_id not in store:
        store[session_id]=ChatMessageHistory()
    return store[session_id]

with_message_history=RunnableWithMessageHistory(model,get_session_history)
```

```
from langchain_core.prompts import ChatPromptTemplate, MessagesPlaceholder
prompt=ChatPromptTemplate.from_messages(
    [
        ("system", "You are a helpful assistant. Answer all the question to the best of your ability"),
        MessagesPlaceholder(variable_name="messages")
    ]
)

chain=prompt|model
```

```
with_message_history=RunnableWithMessageHistory(chain,get_session_history)
```

```
/usr/local/lib/python3.13/dist-packages/IPython/core/interactiveshell.py:3553: LangChainDeprecationWarning: RunnableWithMessageHistory is deprecated. Use RunnableWithMessageHistory instead.
  exec(code_obj, self.user_global_ns, self.user_ns)
```

```
'''config = {"configurable": {"session_id": "chat3"}}
response=with_message_history.invoke(
    [HumanMessage(content="Hi My name is Muragesh")],
    config=config
)

response'''
```

```
'''response = with_message_history.invoke(
    [HumanMessage(content="What's my name?")],
    config=config,
)

response.content'''
```

```
from langchain_core.messages import SystemMessage, HumanMessage, AIMessage, trim_messages
trimmer=trim_messages(
    max_tokens=45,
    strategy="last",
    token_counter=model,
    include_system=True,
    allow_partial=False,
    start_on="human"
)
```

```

messages = [
    SystemMessage(content="you're a good assistant"),
    HumanMessage(content="hi! I'm bob"),
    AIMessage(content="hi!"),
    HumanMessage(content="I like vanilla ice cream"),
    AIMessage(content="nice"),
    HumanMessage(content="whats 2 + 2"),
    AIMessage(content="4"),
    HumanMessage(content="thanks"),
    AIMessage(content="no problem!"),
    HumanMessage(content="having fun?"),
    AIMessage(content="yes!"),
]
trimmer.invoke(messages)

```

```

[SystemMessage(content="you're a good assistant", additional_kwargs={}, response_metadata={}),
HumanMessage(content='whats 2 + 2', additional_kwargs={}, response_metadata={}),
AIMessage(content='4', additional_kwargs={}, response_metadata={}, tool_calls=[], invalid_tool_calls=[]),
HumanMessage(content='thanks', additional_kwargs={}, response_metadata={}),
AIMessage(content='no problem!', additional_kwargs={}, response_metadata={}, tool_calls=[], invalid_tool_calls=[]),
HumanMessage(content='having fun?', additional_kwargs={}, response_metadata={}),
AIMessage(content='yes!', additional_kwargs={}, response_metadata={}, tool_calls=[], invalid_tool_calls=[])]

```

```

from operator import itemgetter

from langchain_core.runnables import RunnablePassthrough

chain=(
    RunnablePassthrough.assign(messages=itemgetter("messages")|trimmer)
    | prompt
    | model
)

response=chain.invoke(
    {
        "messages":messages + [HumanMessage(content="What ice cream do i like")]
    }
)
response.text

```

'I don't know yet-you haven't told me! What's your favorite flavor?'

```

## Lets wrap this in the MMessage History
with_message_history = RunnableWithMessageHistory(
    chain,
    get_session_history,
    input_messages_key="messages",
)
config={"configurable":{"session_id":"chat5"}}

```

```

/usr/local/lib/python3.13/dist-packages/IPython/core/interactiveshell.py:3553: LangChainDeprecationWarning: RunnableWithMessageH
exec(code_obj, self.user_global_ns, self.user_ns)

```

```

response = with_message_history.invoke(
    {
        "messages": messages + [HumanMessage(content="what's my name?")],
    },
    config=config,
)

response.text

```

'Your name is Bob!'

